

A. Table of Notation

In Table 4, we summarize the mathematical notation used throughout the paper, grouped by the track(s) they appear in.

B. Limitations

Our two tracks separate the problems of featurization and causal dependency location for cleaner evaluation. However, these are mutually influential problems: one could potentially locate better circuits by first decomposing MLPs into sparse features, for example (Marks et al., 2025). The featurization method one uses should also be informed by the downstream task; for example, the outputs of DAS are not immediately applicable to finding circuits. Future work should consider the joint problem of (1) building causal dependency graphs from (2) more meaningful units, where these units may potentially exist at various levels of granularity.

For circuit localization, there exist metrics such as completeness that cannot be tractably computed without access to the ground-truth set of causally relevant components. This motivated our inclusion of the InterpBench model, whose AUROC metric includes completeness. Some work has attempted to measure completeness as the faithfulness of the circuit’s complement (Marks et al., 2025), but as it is easy to reduce performance even without ablating the full circuit, this may not be a good signal for when one has recovered the full set of causally relevant dependencies. We acknowledge that a fully automated completeness score is absent for the remaining models.

For causal variable localization, our faithfulness metric captures the extent to which the causal variable—not the entire high-level model—aligns with the representation. The high-level model may differ from the hypothesis, but it would still be possible to modify the model’s behavior in a predictable way, and we believe this will be reflected in the scores. Nonetheless, this paradigm presumes the existence of the high-level model in the computation graph. We have mainly included graphs for which there exists evidence from past work, but we acknowledge that these graphs may not always exist in the models we evaluate in the exact forms shown here.

Finally, our benchmark focuses solely on large language models. Given that this is the current focus of the vast majority of mechanistic interpretability research, we believe that this gives a broad-coverage sample of models commonly studied in the literature. It would be helpful in future work to expand the scope of these evaluations to include other modalities.

C. Further Details on Materials

Table 5 contains dataset statistics.

C.1. Indirect Object Identification (IOI)

To generate IOI examples, we collect sets of templates and attributes—namely, common English first names, common place names, and everyday objects. We separate the templates and attributes into four disjoint groups and use them to generate the four splits (public train/validation/test and private test sets). This means that different splits do not share any attributes. We generate 10,000 IOI examples per split using 43 templates, 166 first names, 319 object names, and 247 place names. The templates and attributes used in the public sets partly overlap with the original dataset by Wang et al. (2023). The rest of the public attributes and the additional private attributes were generated using ChatGPT (OpenAI, 2022) and manually verified. We verify that all names are tokenized to a single token using the test prompt “*I am {name}*” across our models.

In Figure 4, we provide an example from our IOI dataset. Each example includes a prompt, a prompt template, metadata, a list of choices, and the index of the correct completion (answer key) from the list of choices.

Counterfactuals. For each instance, we create eight counterfactuals: the six counterfactuals described by Wang et al. (2023), an additional counterfactual which is the composition of all three transformations proposed by Wang et al. (2023), and a counterfactual where the second instance of the subject is replaced with a third random name that did not appear in the first clause of the sentence (“ABC”). Table 6 contains an example of each counterfactual type.

In the circuit localization track, we use the ABC counterfactual. In the causal variable track, we use the $IO \leftrightarrow S1$ Flip, $IO \leftrightarrow S2$ Flip, and $IO \leftrightarrow S1$ Flip + $IO \leftrightarrow S2$ Flip counterfactuals.

Table 4. The notation used throughout the paper, grouped by track.

Track	Symbol	Meaning
Shared	$\mathcal{N}$	The full computation graph; a neural network
	d_{model}	The size of (i.e., number of neurons in) the output vector of each layer
Circuit Localization	$\mathcal{C}$	A circuit $\in \mathcal{N}$
	k	The proportion of edges in a circuit. If a circuit has $\leq k \times 100$ edges, this is sometimes expressed as $\mathcal{C}_k$
	u	A node in the computation graph. Could be an MLP or an attention head
	(u, v)	An edge from node u to node v
	m	The metric used to evaluate a circuit. Usually the logit difference between a correct and incorrect answer
	f	The faithfulness of a single circuit. Defined as a ratio of m given $\mathcal{C}$ and m given $\mathcal{N}$
	CPR	Circuit performance ratio (higher is better). Defined as the area under the faithfulness curve. An aggregation over f values at many circuit sizes
	CMD	Circuit-model distance (0 is best). Area between the faithfulness curve and 1. An aggregation over f values at many circuit sizes
	N_u	The number of neurons in node u , usually equal to d_{model}
	$N_{\mathcal{C}}$	The set of all neurons in $\mathcal{C}$
Causal Variable Localization	$\mathcal{H}$	A high-level causal model
	X	A variable in $\mathcal{H}$
	b	An base input to the neural network
	c	An counterfactual input to the neural network that differs from b in some systematic way
	$\mathbf{h}$	The output activation vector of a node in the computation graph
	$\mathcal{F}$	The featurization function that transforms $\mathbf{h}$ to a new space where the causal variable X is easier to isolate
	Π_X	A set of “features” in a hidden vector $\mathbf{h}$ abstracted by a variable X , i.e., dimensions in the range of $\mathcal{F}$ that encode X
	$\mathcal{D}$	A dataset containing (b, c) pairs
	$\mathcal{H}_{X \leftarrow \text{get}(\mathcal{H}(c), X)}(b)$	An interchange intervention on the high-level model $\mathcal{H}$ which is run on the input b while the variable X is fixed to the value it takes when $\mathcal{H}$ is run on input c .
	$\mathcal{N}_{\Pi_X \leftarrow \text{get}(\mathcal{N}(c), \Pi_X)}(b)$	A distributed interchange intervention on the LM $\mathcal{N}$ which is run on the input b while the features Π_X of a hidden vector $\mathbf{h}$ are fixed to the value they take when $\mathcal{N}$ is run on input c .

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"prompt": "After Nick and John spent some time
at the car dealership , Nick offered a nail to",
"template": "After {name_A} and {name_B} spent some time
at the {place}, {name_C} offered a {object} to",
"metadata": {
  "indirect_object": "John",
  "subject": "Nick",
  "object": "nail",
  "place": "car dealership",
  "random_a": "Max",
  "random_b": "Fred",
  "random_c": "Bob"
},
"choices": [
  "John",
  "Nick"
],
"answerKey": 0,
"abc_counterfactual": {
  "prompt": "After Nick and John spent some time
at the car dealership , Bob offered a nail to",
  "choices": [
    "John",
    "Nick",
    "Bob"
  ],
  "answerKey": -1
},
"random_names_counterfactual": {
  "prompt": "After Max and Fred spent some time
at the car dealership , Max offered a nail to",
  "choices": [
    "Max",
    "Fred"
  ],
  "answerKey": 1
},
...

```

Figure 4. An IOI example. Each input is paired with a set of templatically generated counterfactuals.